(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FIFTH SEMESTER [B.TECH] JANUARY 2024

Paper Code: CIC-309 ***Subject: Software Engineering***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Use of simple Calculator is allowed.

Q1 Answer **any five** questions:- **(5x3=15)**

(a) Briefly explain the concept of Software Crisis and how it differs from common Software Myths. Provide real-world examples.

(b) Distinguish software re-engineering and reverse engineering?

(c) Calculate and interpret Halstead's effort for a program where the number of distinct operators is n_1=15, the number of distinct operands is n_2=20, the total number of operators isN_1=120, and the total number of operands is N_2=80.

(d) Discuss the implications of high coupling on software maintainability, with examples from industry practices.

(e) Discuss the role of regression testing in Agile methodologies.

(f) Draw the Context level 0 and level 1 DFD for the Safe home Software?

(g) Assess the effectiveness of data dictionaries in managing large-scale software projects, citing real-world applications.

(h) Develop a set of test cases to perform boundary value analysis for the following input fields in an online registration form:

(i) Age: The form requires students to enter their age, which must be between 18 and 30 years inclusive.

(ii) Number of Completed High School Subjects: The form also asks for the number of completed high school subjects, which should be a minimum of 5 and a maximum of 12.

Ensure your test cases cover the boundary conditions for these input fields.

UNIT-I

Q2 (a) Compare and contrast the Waterfall and evolutionary Models of SDLC. Focus on key stages, flexibility, risk management, and suitability for different types of projects. Provide examples. **(7.5)**

(b) Explain the importance and steps of requirement elicitation in software engineering. Discuss the FAST and Use Case approaches, highlighting their strengths and limitations. **(7.5)**

Q3 (a) Describe the key characteristics and structure of an effective Software Requirement Specification (SRS) document. Use IEEE standards as a reference. **(7.5)**

(b) Discuss the differences and similarities between Functional and Non-Functional requirements with examples. How do these impact the software development process? **(7.5)**

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UNIT-II

Q4 (a) Explain the COCOMO cost estimation model. Perform a basic COCOMO calculation for a project with a given size (50 KLOC). Using the Intermediate COCOMO model, calculate the effort in person-months. Assume the project has 'HIGH' complexity [CMPLX=High=1.15]. **(5)**

(b) Describe the concepts of cohesion and coupling in software design. Provide examples of each and discuss their impact on software maintainability and flexibility. **(5)**

(c) Write a short note on live variables and module weakness. **(5)**

Q5 (a) Discuss the Putnam Resource Allocation Model, including its application and limitations. Perform a calculation using given project parameters. **(5)**

(b) Compare and contrast Object-Oriented Design (OOD) and Function-Oriented Design (FOD) in terms of their methodology, advantages, and disadvantages. **(5)**

(c) Differentiate between Function Point and LOC software Metrics. Why Function points metrics are preferred over Lines of Code Metrics. **(5)**

UNIT-III

Q6 (a) Explain Halstead's Software Science metrics. Write equations to calculate the Effort and Time for a given program using Halstead's methos. **(7.5)**

(b) Describe the concept of Software Reliability. Compare the Basic Model and Logarithmic Poisson Model in terms of their approach and applicability. **(7.5)**

Q7 (a) Discuss the importance of software metrics in software engineering. Explain Token Count and Information Flow metrics with examples. **(7.5)**

(b) Explain the CMM (Capability Maturity Model) and ISO 9001 standards. Discuss how they contribute to software quality assurance. Give example of three CMM Level 3 companies based in India. **(7.5)**

UNIT-IV

Q8 (a) Describe the testing process in software engineering, focusing on the importance of functional and structural testing. Provide examples of test cases for each. **(5)**

(b) Create Control Flow Graph for a module that checks if sum of 2 numbers is positive, negative or neutral. Also find cyclomatic complexity number for this module and number of independent paths in the module. **(5)**

(c) Explain the concept and process of Software Maintenance. Discuss the Quick-fix and Boehm's models of maintenance with their merits and demerits. **(5)**

Q9 (a) Define and differentiate between Verification and Validation in software testing. Provide examples to illustrate your points. **(5)**

(b) Discuss the importance and methods of Software Configuration Management in the context of software maintenance. **(5)**

(c) Explain the importance of acceptance testing in software development. Compare and contrast alpha and beta testing. **(5)**

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